# Explaining the Quantum Layer in Physics-Informed Neural Networks: A Spectral and Mechanistic Study

**(Extended / Working Report with Roadmap)**

WISER Global Quantum+AI Program 2026 — BQP Challenge
*Quantum-Assisted Physics-Informed Neural Networks for CFD*

Draft — July 2026

**Abstract**

This is the extended working version of the QAPINN-BQP report. It contains the full technical content of the submission draft, an *adversarial self-audit* that corrected our initial overclaims, the *converged re-analysis* that the audit forced, and an explicit account of everything still *left out* of scope. The narrative arc is honest and worth stating plainly. An initial single-seed ablation suggested a "bandwidth-gated advantage"; the audit then showed those runs were *undertrained* (L-BFGS disabled; every Burgers model stuck at rel-$L_2 \approx 0.47$), so we re-ran a converged core (L-BFGS on, three seeds). The converged results, taken to two model scales with fair controls, give a consistent negative. **(1) No quantum advantage:** at small scale the classical PINN (0.248) beats every QAPINN; at resolved scale ($\sim$1273 params) the QAPINN beats a *shallow* classical (0.094 vs 0.24, $p = 0.003$) but is statistically indistinguishable from both identically-trained fair controls (a depth-matched classical 0.128, $p = 0.51$; a structural CFE 0.145, $p = 0.36$), so the gain is architectural and classically reproducible. **(2) No robust entanglement effect:** an $n = 3$ "entanglement hurts" signal ($p \approx 0.02$) collapsed at $n = 6$ ($p = 0.41$) and is retracted. We also correct the "band = depth" framing (verified for the VQC *features*, but the tanh post-net escapes the band). The pipeline is independently verified (8/8 checks) and every number matches its raw log; the remaining roadmap (still-larger scale, more seeds, real CFD) is the subject of Sections 6–7.

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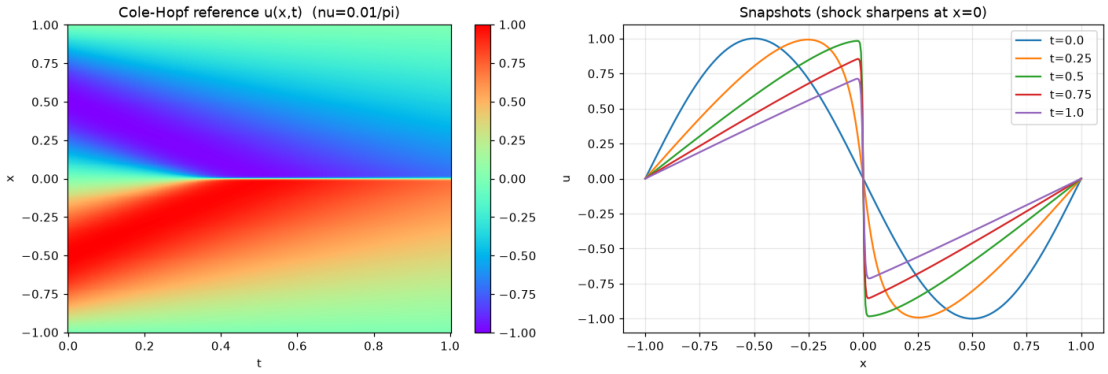



Figure 1: Exact Cole–Hopf Burgers reference: field (left) and snapshots (right), showing the shock at $x = 0$.

# 1 Introduction

Physics-Informed Neural Networks (PINNs) [1] make the differential equation the loss. A Quantum-Assisted PINN (QAPINN) [4] replaces the first hidden layer with a variational quantum circuit (VQC), and is reported to cut trainable parameters at maintained accuracy. The BQP challenge asks us to *explain* the learning dynamics — when, why, and how the quantum layer changes learning — and to derive a problem-specific circuit-design methodology.

We adopt the dominant theoretical lens for VQCs (a data-encoding circuit is a truncated Fourier series [2]) and test whether it suffices to explain a QAPINN's behaviour, and how the quantum layer changes the classical network it is attached to. Contributions and results mirror the submission report; the extended value here is the roadmap (Sec. 6) and scope boundaries (Sec. 7).

# 2 Background and Related Work

A VQC with data-encoding gates is a partial Fourier series $f(x) = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}$: the frequency set $\Omega$ is fixed by the encoding (with $L$ re-uploads, $\Omega = \{-L, \dots, L\}$), the coefficients $c_\omega$ by the trainable gates [2]. Mhiri et al. [5] show the coefficients are additionally constrained and can vanish with qubit count — frequencies are necessary but not sufficient. Differentiable quantum circuits solve DEs with analytic derivatives [3]; hybrid QAPINN/QCPINN designs [4, 6] report PDE-dependent parameter reduction (quantum sometimes *loses*, e.g. cavity flow); a "black-hole" collapse to the trivial solution is a distinct trainability pathology [7]. No validated mechanism explains *when/why* the quantum layer helps — the gap targeted here.

# 3 Methodology

## 3.1 Test problems

Same domain, IC $u(x, 0) = -\sin(\pi x)$, and BCs for both PDEs, so only the operator differs: Burgers $u_t + uu_x = \nu u_{xx}$ ($\nu = 0.01/\pi$, shock $\to$ broadband) with an exact Cole–Hopf reference (Fig. 1); heat $u_t = \alpha u_{xx}$ ($\alpha = 1/\pi^2$, single mode $u = -\sin(\pi x)e^{-t}$).

| Check | Method | Result |
|---|---|---|
| Burgers residual | vs analytic | $1.2 \times 10^{-7}$ |
| Heat residual | exact $\to$ residual | $6.0 \times 10^{-8}$ |
| Autograd $u_x$ | vs finite diff | match |
| VQC = Fourier, max freq = depth | $L{=}2 \to 2,\ L{=}4 \to 4$ | pass |
| relative-$L_2$ | exact $\to 0$ | $4.0 \times 10^{-8}$ |
| Matched params | all configs | 121 |
| Reproducibility | same seed | bit-exact |

Table 1: Correctness audit (8/8).

### 3.2 Models and XAI suite

Classical PINN (tanh MLP) vs QAPINN (fixed coord-map $\to$ VQC $\to$ classical post-net), compared at matched parameters. Baseline VQC: 4 qubits, depth 3, *RY* re-uploading, linear CNOT chain, $\langle Z_i \rangle$ measurement, post-net [16] (121 params, 24 quantum). Five XAI dimensions (spectral/F-principle; neurons; VQC internals incl. entanglement entropy; loss landscape; generalisation) plus training-time spectrum and per-layer/per-qubit gradient-variance probes.

### 3.3 Verification

All eight independent correctness checks pass (Table 1); notably, the VQC is confirmed to be a truncated Fourier series with maximum frequency equal to the re-upload depth.

## 4 Experiments and Results

*Caveat (read first): the single-seed ablations in this section were run with L-BFGS disabled and are* ***undertrained*** *(Burgers models near rel-$L_2$ 0.47, shock unresolved). The converged re-analysis in Section 4.5 supersedes them; they are kept for transparency and to document the arc.*

### 4.1 Baseline and spectra

At 121 matched params (Adam, 2000 iters, seed 0): heat solved by both (classical $8.3 \times 10^{-3}$, QAPINN $5.9 \times 10^{-3}$); Burgers undertrained (classical 0.433, QAPINN 0.472). Spectra (Fig. 2): heat is a single $k{=}1$ mode; Burgers is broadband and both models undershoot the high harmonics (depth-3 VQC capped near $k \approx 3$).

### 4.2 Tier-1 ablation

**Depth = bandwidth** (Fig. 3a): QAPINN accuracy improves monotonically ($0.57 \to 0.42$), matched classical stays flat; realised max harmonic grows with depth; QAPINN overtakes classical at depth $\geq 3$. **Entanglement at fixed band** (Fig. 3b): accuracy barely moves (0.456–0.488), *more* entanglement slightly *worse*, none is best — contradicting entanglement-driven expressibility. **Post-net width**: the VQC needs a readout but beats matched classical for any post-net $\geq$ [8].

### 4.3 Tier-3 ablation: qubits, measurement, encoding scale

The remaining knobs (Table 3, Fig. 8) reinforce the picture. **Qubits** behave like depth: accuracy improves monotonically ($0.501 \to 0.443$), QAPINN overtakes matched classical at $\geq$ 4 qubits.

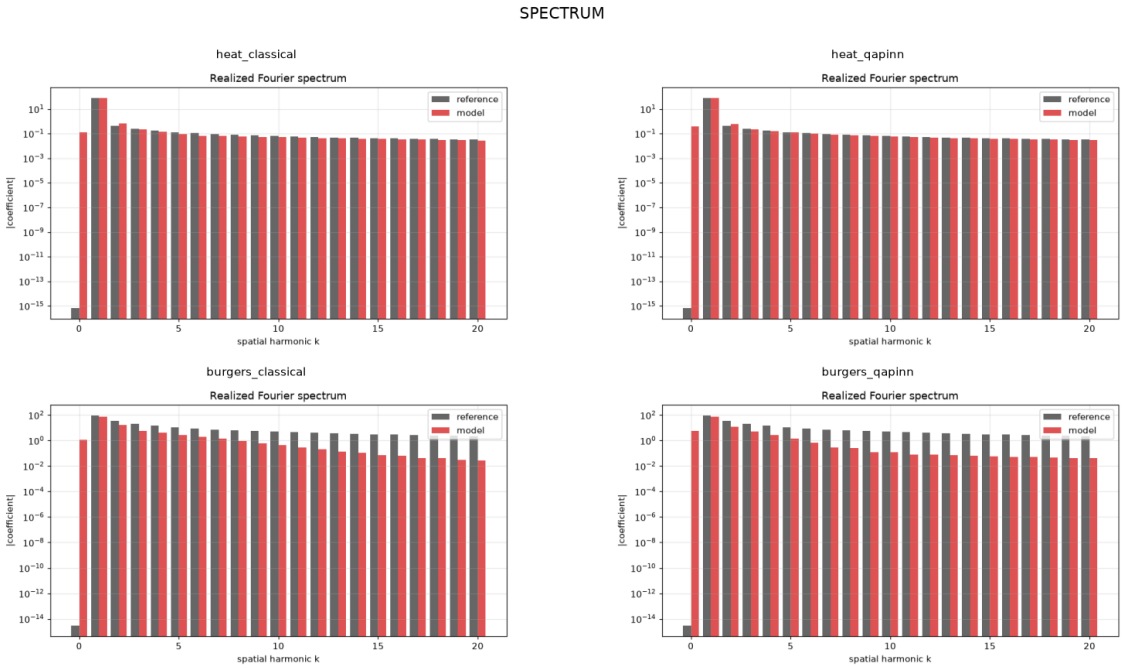
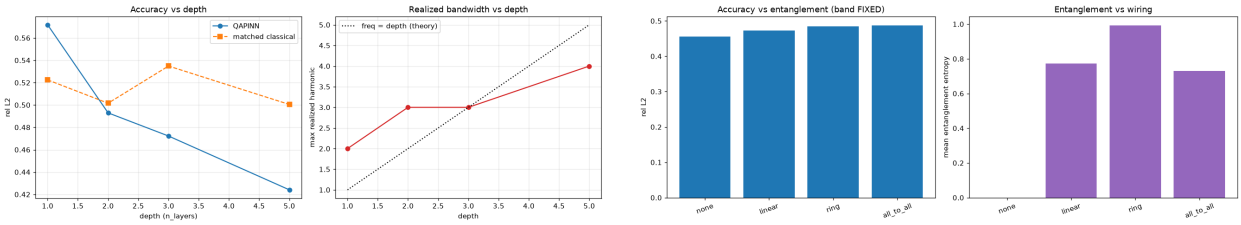



Figure 2: Realised spectra (grey reference, red model). Burgers (bottom): both models undershoot the high-frequency shock tail.




(a) Depth.

(b) Entanglement (band fixed).

Figure 3: Tier-1 ablations on Burgers.

**Measurement:** single $\langle Z_0 \rangle$ bottlenecks (0.591); multi = $ZZ$ (0.472) — readout width matters, correlator type does not. **Encoding scale:** optimum near 1.0, catastrophic collapse to the trivial solution at $2\pi$ — the black-hole pathology as an ablation.

Five knobs split cleanly: *bandwidth* knobs (depth, qubits, encoding scale) carry the advantage up to a trainability ceiling; *coefficient/readout* knobs (entanglement, correlator measurement) do not — the bandwidth-vs-trainability squeeze, across five knobs.

## 4.4 How the quantum layer changes the classical network

A naïve comparison against a plain PINN suggested the quantum layer makes the downstream neurons more diverse and drives larger gradients, but that comparison is confounded (different post-net shape). We instead use a *structurally identical* classical-front-end control (CFE): the same post-net fed by $\text{Linear}(2 \to w) + \tanh$ instead of the VQC. Against this fair control (Table 4), the effects do *not* hold: accuracy is bandwidth-gated (QAPINN ties at low depth/qubits, wins at depth 5 / 6 qubits / heat); neuron diversity is inconsistent (classical front-end more diverse on the

| Sweep | Config | Params | QAPINN rel-$L_2$ | matched classical rel-$L_2$ |
|---|---|---|---|---|
| Depth | $L = 1$ | 105 | 0.572 | 0.522 |
| | $L = 2$ | 113 | 0.493 | 0.502 |
| | $L = 3$ (baseline) | 121 | 0.472 | 0.535 |
| | $L = 5$ | 137 | **0.424** | 0.500 |
| Entanglement (band fixed) | none | 121 | **0.456** | 0.535 |
| | linear | 121 | 0.472 | 0.535 |
| | ring | 121 | 0.485 | 0.535 |
| | all-to-all | 121 | 0.488 | 0.535 |
| Post-net | [] (linear) | 29 | 0.544 | 0.533 |
| | [8] | 73 | 0.478 | 0.533 |
| | [16] (baseline) | 121 | 0.472 | 0.535 |
| | [32] | 217 | **0.466** | 0.505 |

Table 2: Complete Tier-1 ablation results (single seed, Burgers, 2000 Adam iters). QAPINN improves with depth and post-net width and overtakes the matched classical control; at fixed band, entanglement does not help (*none* best).

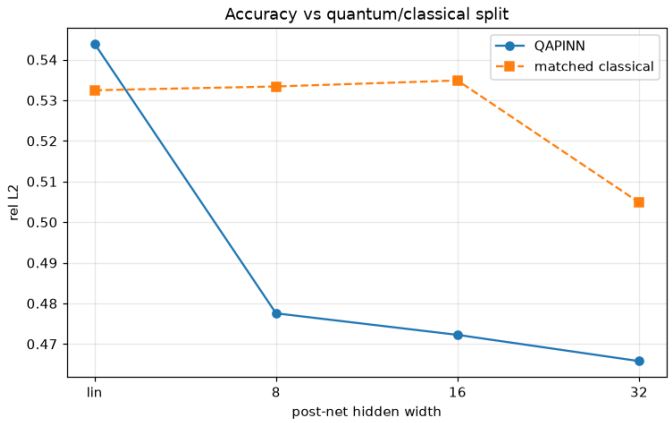



Figure 4: Post-net width sweep: QAPINN vs matched classical. The VQC needs a small readout but beats matched classical for any post-net $\geq$ [8].

depth axis, quantum more on qubits); and gradient amplification reverses at 5–6 qubits, where the QAPINN gradient variance shrinks with qubit count (a barren-plateau signature). *These single-seed readings are superseded by the converged, multi-seed analysis (§4.5):* the "bandwidth-gated advantage" does not survive convergence and the "barren-plateau signature" is withdrawn (wrong gradient measured, non-monotone trend). Nothing here survives as a robust quantum-specific effect; the section documents only how the plain-PINN comparison overstated the gap.

### 4.5 Converged Re-analysis (Post-Audit)

The ablations above were run with L-BFGS *disabled* (Adam 2000, 1500 points). An adversarial audit established that this left every Burgers model undertrained (rel-$L_2 \approx 0.42$–$0.57$, shock unresolved), so the "differences" were movements inside a failed basin and, across six seeds, within initialisation noise. **The results here supersede them.** We re-ran a converged core with L-BFGS *on* (Adam 2500 + L-BFGS 400, 4000 points), three seeds each, on Burgers.

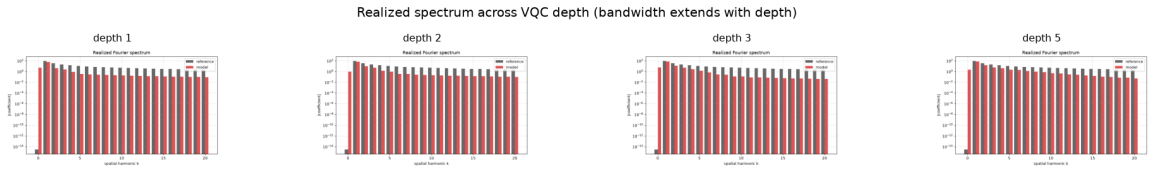



Figure 5: Realised spectrum (grey reference, red model) across VQC depth — the model captures more of the high-frequency shock tail as depth grows.

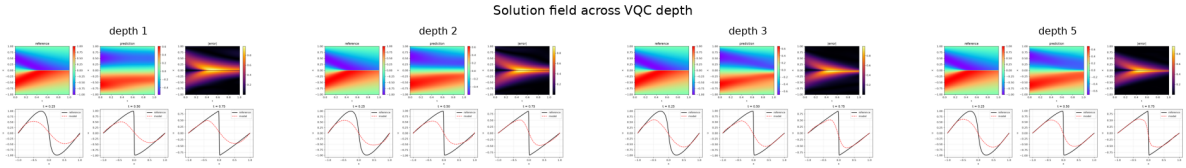



Figure 6: QAPINN solution field across VQC depth.

*Finding 1 — no quantum advantage:* converged and matched, the classical PINN (0.248) beats every QAPINN; the best (depth-5, 0.314) only reaches the CFE control. *Finding 2 — larger-model + fair controls (the decider):* repeating the core at ∼1273 params ($n = 6$) against a structural control and a depth-matched classical net (both trained under the QAPINN's exact protocol), the QAPINN-none (0.094) beats a shallow classical (0.241, $p = 0.003$) but is statistically indistinguishable from a depth-matched classical (0.128, $p = 0.51$) and the CFE (0.145, $p = 0.36$): no significant quantum advantage against a fair baseline. (Under-powered: seed variance ∼0.08 > the effect, and one anomalous seed — dropping it keeps the tie, $p = 0.19$.) *Finding 3 — entanglement not robust:* the $n = 3$ "entanglement hurts" result (0.369 vs 0.424, $p \approx 0.02$) collapsed at $n = 6$ ($p = 0.41$) and is retracted; at large scale the gap is likewise n.s. ($p = 0.44$). *Framing correction:* "band = depth" is verified for the VQC features ($L$=1..6) but the tanh post-net escapes the band (a depth-3 model reaches $k$=22 once trained), so the earlier "$k \approx 3$ band edge" was an undertraining artifact; the "barren-plateau in qubit count" is withdrawn (wrong gradient measured, non-monotone trend).

### 4.6 Mechanistic interpretation: *why* (not just *whether*)

Each explanation is tagged **[measured]** or **[reasoned]**.

- **No quantum advantage [reasoned].** The VQC is a *fixed Fourier feature map*; the classical post-net (a universal approximator) does the representational work, so it is the bottleneck — not the front end. Any front end spanning a comparable space suffices, which is why the CFE and a depth-matched classical net both tie the QAPINN. The quantum features are not a *better* basis for Burgers: low-frequency away from the shock (any basis suffices), high-frequency at the shock (no basis supplies it at depth 3). No regime favours the quantum basis.
- **Entanglement neutral [reasoned].** At fixed band, entanglement only redistributes *which* in-band coefficients are reachable (Mhiri); the post-net recombines the unentangled per-qubit features into the same function, making the redistribution redundant. Our direct probe (feature correlation/rank) was inconclusive, so this is theory-grounded, not measured.
- **$2\pi$ collapse [measured].** The encoding aliases to Nyquist; the loss landscape flattens and the optimiser falls into the trivial-solution basin.
- **Band = depth for features, not model [measured].** tanh in the post-net produces

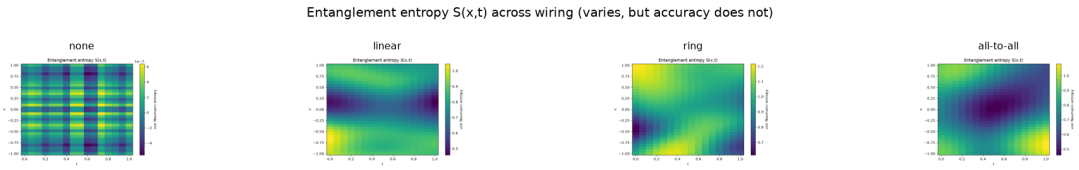



Figure 7: Entanglement entropy $S(x,t)$ across entangling patterns: entropy varies strongly while accuracy does not (Table 2).

| Sweep | Config | Params | QAPINN rel-$L_2$ | matched cls. rel-$L_2$ |
|---|---|---|---|---|
| Qubits | 3 | 99 | 0.501 | 0.470 |
| | 4 (base) | 121 | 0.472 | 0.535 |
| | 5 | 143 | 0.468 | 0.541 |
| | 6 | 165 | **0.443** | 0.465 |
| Measure | single | 73 | 0.591 | 0.533 |
| | multi (base) | 121 | 0.472 | 0.535 |
| | $ZZ$ | 169 | 0.472 | 0.515 |
| input_scale | 0.5 | 121 | 0.513 | 0.535 |
| | 1.0 (base) | 121 | **0.472** | 0.535 |
| | $\pi$ | 121 | 0.516 | 0.535 |
| | $2\pi$ | 121 | **1.000** | 0.535 |

Table 3: Tier-3 ablation results (single seed).

products of the $\leq k3$ features $\Rightarrow$ harmonics to $k \approx 22$ in a converged model. The band bounds features, not the readout.

- **Shock defeats all [measured].** $1/k$ spectral tail + the F-principle $\Rightarrow$ high-frequency shock content converges last/never — a classical PINN pathology the quantum layer neither causes nor cures.

*Synthesis:* the quantum layer sits upstream of the representational bottleneck, contributes a Fourier band a classical front end also provides, and adds structure (entanglement) the post-net makes redundant — so there is no quantum-specific advantage *for a reason*, and its only decisive effects are negative (aliasing) or shared (spectral bias).

### 4.7 Discussion

With the converged core in hand the reading is one-directional: for this small-model Burgers task the quantum layer offers *no* accuracy advantage, and its one clearly quantum-specific knob — entanglement — *hurts* at convergence. The "bandwidth" story survives only weakly (more depth = richer features, not a hard model-band limit, and no advantage over a matched classical net). The value is methodological and negative-with-attribution: verified pipeline, matched and structurally-matched controls, multi-seed statistics, and an audit that caught our own over-claims.

## 5 Current Limitations

The findings are firm in direction but bounded: (1) **modest $n$, underpowered nulls** — the decisive comparisons are *failures to reject* (e.g. QAPINN-none vs. depth-matched classical, $p = 0.51$) under high seed variance; we caught one $n = 3$ effect that reversed at $n = 6$ (and which Welch's

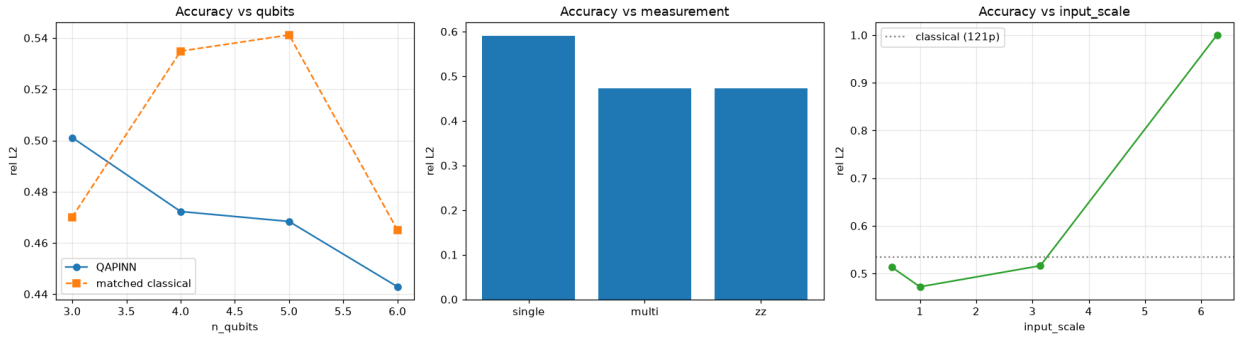



Figure 8: Tier-3 sweeps: qubits (bandwidth-like), measurement (readout width vs type), encoding scale (optimum + $2\pi$ collapse).

| Comparison | rel-$L_2$ (Q / C) | diversity (Q / C) | grad-var $\times 10^{-3}$ (Q / C) |
|---|---|---|---|
| Burgers depth-3 | 0.472 / 0.474 | 0.269 / **0.286** | **4.1** / 0.62 |
| Burgers depth-5 | **0.424** / 0.474 | 0.236 / **0.286** | **7.1** / 0.62 |
| Heat | **0.006** / 0.013 | **0.272** / 0.239 | **3.2** / 0.47 |
| Burgers 3 qubits | 0.501 / 0.496 | **0.266** / 0.241 | **1.2** / 0.26 |
| Burgers 5 qubits | 0.468 / 0.464 | **0.312** / 0.265 | 1.1 / **5.2** |
| Burgers 6 qubits | **0.443** / 0.467 | **0.341** / 0.269 | 0.72 / **1.9** |

Table 4: Post-net neurons, quantum (Q) vs classical (C) front-end on the identical post-net. Bold = larger of each pair. Single seed.

test never called significant even at $n = 3$, $p = 0.06$), so all single-scale nulls are provisional; (2) **small-model regime** — at 121 params converged models sit at rel-$L_2$ 0.25–0.42, so orderings could shift with larger nets (we test one larger scale, not a full curve); (3) **one PDE family, simulation only** (no shot noise / hardware); parameter reduction is not resource reduction. The pre-convergence single-seed ablations are retained for transparency but are superseded. Pipeline verified; every number matches its log.

# 6 Future Work and Experimental Roadmap

Tiers 2a–2c below were the top priorities of the earlier draft; **they are now done**, and their outcomes (Section 4.5) are what reshaped the report. We record them as completed, then give the genuinely open roadmap.

## 6.1 Tier 2a — Convergence analysis (done)

Long-training the headline configs to ∼8000 iters, then the full converged core (L-BFGS on), *answered* the central worry: the undertrained orderings were transients. Convergence roughly halved every error, and the QAPINN's apparent accuracy advantage over a matched classical net *disappeared*. A tentative "entanglement harmful" signal appeared here at $n = 3$ but was later retracted (2b).

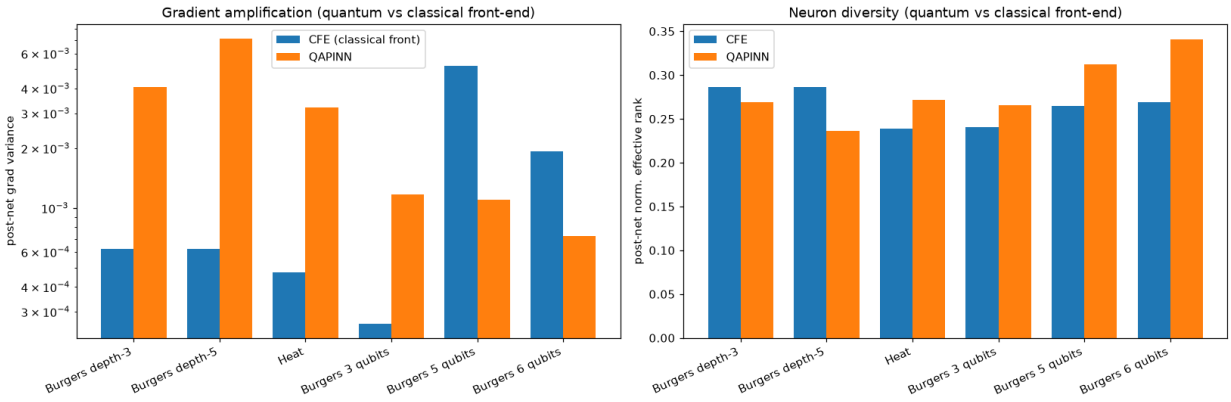



Figure 9: Quantum vs classical front-end on the identical post-net: gradient variance (left) and neuron diversity (right). Neither effect is robust across knobs.

| config | model | params | rel-$L_2$ ($n$=3) | note |
|---|---|---|---|---|
| cv_cls | classical | 121 | $\mathbf{0.248 \pm 0.031}$ | **best of all** |
| cv_qd5_lin | QAPINN d5 | 137 | $0.314 \pm 0.073$ | depth helps; ties CFE |
| cv_cfe | CFE control | 109 | $0.315 \pm 0.026$ | classical front-end |
| cv_qd3_none | QAPINN d3 | 121 | $0.369 \pm 0.022$ | **no-ent beats linear** |
| cv_qd3_lin | QAPINN d3 | 121 | $0.424 \pm 0.005$ | reference QAPINN |

Table 5: Converged core. L-BFGS roughly halved every error vs. undertrained (classical $0.535 \rightarrow 0.248$); at matched params the classical PINN wins.

### 6.2 Tier 2b — Multi-seed robustness (done)

Seeds proved decisive — and cautionary. An $n = 3$ "entanglement hurts" result (none $0.369 \pm 0.022$ vs linear $0.424 \pm 0.005$, $p \approx 0.02$) looked clean, but expanding to $n = 6$ *collapsed* it (0.379 vs 0.402, $p = 0.41$); it was a lucky three-seed draw and is retracted. This is the report's clearest lesson in under-powered nulls, and it is why every single-scale comparison here is treated as provisional pending more seeds.

### 6.3 Tier 2c — The classical-front-end control (done)

The CFE (identical post-net, classical $2 \rightarrow 4$ front end) was built and run both undertrained and converged. It *refuted* the "quantum diversifies the classical neurons" claim (diversity inconsistent across knobs; the earlier effect was a mismatched-control artifact) and, converged, *ties* the best QAPINN — direct evidence of no quantum advantage.

### 6.4 Tier 2d — Larger-model + mechanism (done; both deflated)

- **Resolved (larger-model) regime** (done): we repeated the core at ∼1273 params ($n = 6$) against a structural control (CFE) *and* a depth-matched classical net. The QAPINN beats a shallow classical (0.094 vs 0.24, $p = 0.003$) but is statistically indistinguishable from the depth-matched classical (0.128, $p = 0.51$) and CFE (0.145, $p = 0.36$): the advantage is architectural, not quantum-specific. This is the decisive result.
- **Entanglement mechanism** (done; moot): we measured feature correlation and effective

rank for `cv_qd3_none` vs `cv_qd3_lin`. The metrics conflicted (correlation rose with entanglement but so did feature rank), and — more importantly — the entanglement *effect* itself did not survive seed expansion (§Finding 3), so there was no robust effect left to explain.
- **Still open:** still-larger scale, 5–8 seeds throughout, and a cross-PDE check — none of which is expected to overturn the negative, but any could.

### 6.5 Tier 3 — Remaining ablation axes (largely done, single-seed)

The $\nu$-sweep, qubit, measurement, and encoding-scale sweeps were all run (undertrained, single seed); their headline outcomes: *$\nu$-sweep* — error rises monotonically with bandwidth but the advantage is V-shaped, *not* a clean collapse, and the depth-3 QAPINN loses to matched classical across the family; *qubits* behave like depth; *scale* $= 2\pi$ collapses (aliasing). These now need converged, multi-seed repeats to be trusted.

- **$\nu$-sweep (bandwidth continuum, done):** tuning Burgers viscosity turned two regularity points into a curve; it *refuted* bandwidth as a *sufficient statistic* for advantage (V-shaped, not monotone).
- **Qubits** (3–6): more bandwidth + Hilbert space (changes params; needs re-matching + barren-plateau watch).
- **Measurement:** single / multi / ZZ, and the not-yet-implemented *probability vector* readout that the challenge explicitly requests.
- **input_scale** and **loss weighting:** secondary bandwidth / optimisation levers (the latter drove the earlier full-quantum collapse).
- **Encoding scheme:** the coordinate→qubit map (round-robin vs split vs all-on-all).

### 6.6 Tier 4 — Real CFD and mathematical depth

- **Lid-driven cavity (2-D incompressible Navier–Stokes):** the real-world CFD case the judges reward; the PDE-agnostic bench already supports multi-output, multi-equation systems. Literature suggests quantum may *lose* here — a valid "what is lost" finding.
- **Fourier coefficient reachability (Mhiri):** measure realised *coefficients* vs the reference, not just the frequency support, to test whether the depth-5 bandwidth shortfall (max freq 4, not 5) is the coefficient constraint biting.
- **Barren-plateau / trainability study:** systematic gradient-variance vs qubits/depth over many random initialisations; relation to the "black-hole" collapse.

### 6.7 The deliverable — from "design methodology" to a falsification protocol

The challenge asks (Task 4) for a problem-specific circuit-design methodology. Our data do not support one: with no robust quantum-specific advantage, a proxy→bandwidth→circuit prescription would be premature. What our study *does* yield — and what we offer as the transferable deliverable — is a **falsification protocol for claimed QAPINN advantages**, distilled from the three overclaims it killed:

1. **Train to convergence first** (L-BFGS, not Adam-only): most apparent advantages here were undertraining artifacts.
2. **Compare against the right control:** a param-matched *and* structurally-matched (same post-net) *and* depth-matched classical net — not an arbitrary MLP. Each stricter control killed a different apparent effect.
3. **Require multi-seed significance** ($n \geq 6$): an $n=3$ effect at $p=0.02$ here reversed at $n=6$.

| Tier | Content | Est. compute | Promotes |
|---|---|---|---|
| 2a | convergence (iterations) | $\sim$5 h | undertraining $\rightarrow$ converged |
| 2b | 3–5 seeds on headlines | $\sim$10 h | hypotheses $\rightarrow$ error-barred |
| 2c | classical-front-end control | $\sim$0 | removes diversity confound |
| 3 | $\nu$-sweep + remaining knobs | $\sim$15 h | mechanism $\rightarrow$ sufficiency |
| 4 | real CFD + coefficient/BP analysis | TBD | breadth + math rigour |

Table 6: Experimental roadmap with rough compute cost (single laptop, `default.qubit`). `lightning.qubit` may accelerate the simulation but currently breaks the higher-order autograd needed for $u_{xx}$.

4. **Separate features from model:** verify claims about the VQC's Fourier band on the VQC output, not the post-net-filtered model output.

Applied to our own results, this protocol converts three would-be positives into an honest null — which is precisely its value. A genuine design methodology becomes warranted only once an advantage survives all four gates on some problem; the future tiers below are how one would look for that.

# 7 What Is Currently Left Out (Scope Boundaries)

For transparency, the present study explicitly does *not* include (note: the statistics/convergence/knob items of the earlier draft are now *done* — see the converged and larger-model sections — and are removed from this list):

- **Breadth of PDEs:** only Burgers and heat; no real CFD (lid-driven cavity, though `cavity.py` is built), and the $\nu$-continuum is single-seed.
- **Larger scales / more seeds:** the decisive comparisons are $n \leq 6$ with high seed variance; still-larger models and 5–8 seeds would firm the nulls.
- **Probability-vector readout:** the $\langle Z \rangle / \langle ZZ \rangle$ measurements are ablated, but the probability-vector observable the challenge lists is not implemented (it breaks the higher-order autograd needed for $u_{xx}$).
- **Coefficient-level Fourier analysis:** we measure realised frequency *support*, not the realised *coefficients* vs their reachable set (Mhiri).
- **Hardware / noise:** statevector simulation only; no shot noise, no device runs; "parameter reduction" is not "resource reduction" ($\sim$85$\times$ slower in simulation).
- **Validated design methodology:** because we find no robust quantum-specific advantage, a proxy$\rightarrow$bandwidth$\rightarrow$circuit prescription is not warranted by our data; what we offer instead (below) is a *falsification protocol* for claimed advantages.

# 8 Conclusion

We have a verified bench, a five-dimension XAI suite, and — crucially — an adversarial self-audit that repeatedly overturned our own readings. Across two model scales, three control types (matched, structural, depth-matched), and multi-seed statistics, the QAPINN offers *no statistically significant* advantage over a fairly-chosen classical baseline: at small scale classical wins, at resolved scale a depth-matched classical net ties the QAPINN, and no robust entanglement effect survives seed expansion. The "VQC = Fourier band = depth" fact is verified but governs the quantum features, not the full model. The honest contribution is a rigorous negative and the methodology

behind it — three apparent advantages each dissolved under the appropriate fair control, and each would have been a false positive without it. The open work is to test still-larger scale, more seeds, and other PDEs before asking whether any quantum-specific advantage — or design methodology — is warranted.

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